Chapter 5: Convergence Tests

Purpose of Convergence Test

The idea of convergence test is to determine the convergence/divergence of as a series **without** working out explicitly the corresponding limit of partial sums.

So convergence tests only provide a **Yes/No** answer, and often take the form:

Certain test criteria is satisfied $\Longrightarrow$ we have convergence/divergence

5.1 The Integral Test

Reference: Stewart's Calculus 8E, Section 11.3

Goal: Determine the convergence of a series using the Integral Test. An important example is the convergence/divergence of a p-series.

A Motivating Example:

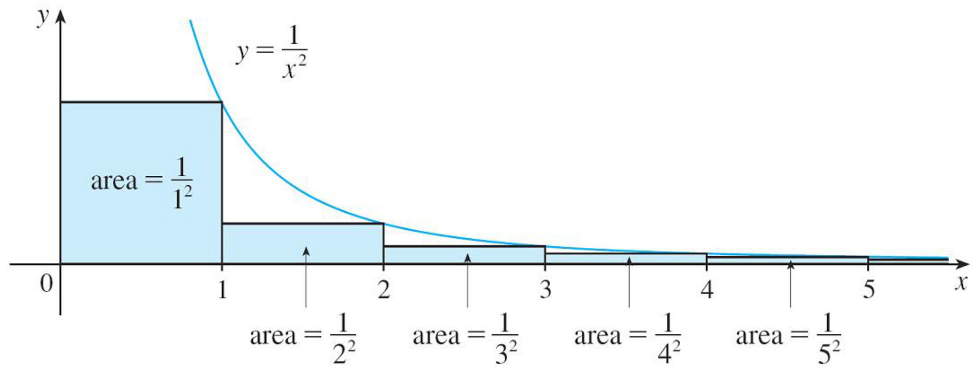
Consider the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$.

It is known that the improper integral $\displaystyle\int_1^{\infty} \frac{1}{x^2}\,dx$ is convergent, since

$$\int_1^{\infty} \frac{1}{x^2}\,dx = \lim_{t\to\infty} \left[\frac{x^{-1}}{-1}\right]_1^t = 1.$$

Note that

$$\frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \leq \int_1^{\infty} \frac{1}{x^2}\,dx = 1.$$



Therefore, the k-th partial sums of the series is bounded:

$$s_k = \frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{k^2} \leq 1 + \int_1^\infty \frac{1}{x^2}\, dx = 1 + 1 = 2.$$

→ as long as all the terms in the series are positive, the partial sum is always increasing.

Also, the sequence of partial sums $\{s_k\}$ is increasing, since $s_{k+1} = s_k + \overbrace{\frac{1}{(k+1)^2}}^{a_{k+1}} > s_k$.

Hence, by the Monotonic Sequence Theorem (Chapter 4, Theorem 5), the sequence $\{s_k\}$ is convergent, i.e. the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Theorem 1. (The Integral Test)

→ to integrate → sequence of partial sum is always increasing → so that integral has a chance to converge.

Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ (N), and let $a_n = f(n)$.

(i) If $\int_1^\infty f(x)\, dx$ (N) is convergent, then the series $\sum_{n=1}^\infty a_n$ is convergent.

(ii) If $\int_1^\infty f(x)\, dx$ (N) is divergent, then $\sum_{n=1}^\infty a_n$ is divergent.

NOTE: When using the Integral Test, it is not necessary that the series or the integral starts at $n = 1$. For example, in testing the series $\displaystyle\sum_{n=4}^{\infty} \frac{1}{(n-3)^2}$, we can use the integral $\displaystyle\int_4^\infty \frac{1}{(x-3)^2}\, dx$.

NOTE: It is not necessary that f is always decreasing. What is important is that f is *ultimately* decreasing, that is f is decreasing for all $x \geq N$ for some number N.

Example 5.1. Test the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{1+n^2}$ for convergence.

Solution. The function $f(x) = \dfrac{1}{1+x^2}$ is continuous, positive and decreasing on $[1, \infty)$.

$$\begin{aligned}
\int_1^{\infty} \frac{1}{1+x^2}\,dx &= \lim_{t\to\infty} \int_1^t \frac{1}{1+x^2}\,dx \\
&= \lim_{t\to\infty} \left[\tan^{-1} x\right]_1^t \\
&= \lim_{t\to\infty} \left(\tan^{-1} t - \frac{\pi}{4}\right) \\
&= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}.
\end{aligned}$$

$$\int_1^{\infty} \frac{1}{1+x^2}\,dx \le \int_1^{\infty} \frac{1}{x^2}\,dx = \left[-\frac{1}{x}\right]_1^{\infty} = -\frac{1}{\infty} - (-1)$$

$$\lim_{t\to\infty} \int_1^t \frac{1}{x^2}\,dx = \lim_{t\to\infty} \left[-\frac{1}{x}\right]_1^t = -0 - (-1) = 1$$

↳ also can conclude convergence, we don't need to actual value it converges to, we just checking if it converges

By the Integral Test, the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{1+n^2}$ is convergent.

An important class of series consists of p-series. The convergence/divergence of p-series can be determined using the Integral Test.

Theorem 2. **(p-series)**

A p-series is a series of the form $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ where p is some fixed real number. A p-series is

- convergent if $p > 1$;
- divergent if $p \le 1$.

Proof. We consider different cases depending on p.

- $\boxed{p < 0}$

  $= n^{-p}$ → positive

  Then the term $\dfrac{1}{n^p}$ can be arbitrarily large for sufficiently large n, i.e. $\displaystyle\lim_{n\to\infty} \frac{1}{n^p} = \infty$. So, by the n-th Term Test of Divergence (Chapter 4, Theorem 8), the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent.

  $a_n \not\to 0 \Rightarrow \sum a_n$ diverges

- $\boxed{p = 0}$

  Then the term $\frac{1}{n^p} = 1$ for all 1, and so $\lim_{n\to\infty} \frac{1}{n^p} = 1 \neq 0$. Again, by the n-th Term Test of Divergence (Chapter 4, Theorem 8), the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent.

- $\boxed{p = 1}$

  Then $\frac{1}{n^p} = \frac{1}{n}$. So the p-series becomes the harmonic series which is known to be divergent (see Example 4.24).

  $f(x) = \frac{1}{x^p}$

  $f'(x) = -px^{-p-1} < 0 \ \forall x$

  $\therefore$ decreasing on $[1, \infty)$

- $\boxed{p \neq 1, \text{ and } p > 0}$ $\to p>1$ $\to 0<p<1$

  Then

$$\begin{aligned}
\int_1^\infty \frac{1}{x^p}\,dx &= \lim_{t\to\infty} \int_1^t x^{-p}\,dx \\
&= \lim_{t\to\infty} \left[\frac{x^{-p+1}}{-p+1}\right]_{x=1}^{x=t} \\
&= \lim_{t\to\infty} \frac{1}{1-p}\left(t^{1-p} - 1\right)
\end{aligned}$$

  Consider the following cases:

  (i) If $p > 1$, then $\underset{1-p<0}{p - 1 > 0}$, and so $t^{1-p} = \frac{1}{t^{p-1}} \to 0$ as $t \to \infty$. Therefore,

$$\int_1^\infty \frac{1}{x^p}\,dx = \lim_{t\to\infty} \frac{1}{1-p}\left(t^{1-p} - 1\right) = \frac{1}{1-p}(0-1) = \frac{1}{p-1}.$$

  $\to$ only time it converges.

  By the Integral Test, the corresponding p-series is convergent.

  (ii) If $0 < p < 1$, then $1 - p > 0$, and so $t^{1-p} \to \infty$ as $t \to \infty$. Therefore,

$$\int_1^\infty \frac{1}{x^p}\,dx = \lim_{t\to\infty} \frac{1}{1-p}\left(t^{1-p} - 1\right) = \infty. \quad \text{(divergent)}$$

  By the Integral Test, the corresponding p-series is divergent.

$\square$

Example 5.2. Determine whether the series $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ converges or diverges.

<u>*Solution.*</u> Consider the function $f(x) = \dfrac{\ln x}{x}$. Clearly, $f(x)$ is positive and continuous for all $x > 1$. To show that f is decreasing, we compute the derivative

$$f'(x) = \frac{(1/x)x - \ln x}{x^2} = \frac{1 - \ln x}{x^2}.$$

Thus, $f'(x) < 0$ for all $\ln x > 1$, i.e. when $x > e = 2.718\ldots$. Therefore, f is decreasing for all $x \geq 3$.

In view of the Integral Test, we now check the improper integral:

$$\begin{aligned}
\int_3^\infty \frac{\ln x}{x}\,dx &= \lim_{t\to\infty} \int_3^t \frac{\ln x}{x}\,dx \\
&= \lim_{t\to\infty} \left[\frac{1}{2}(\ln x)^2\right]_3^t \quad \text{(integration by parts)} \\
&= \lim_{t\to\infty} \left(\frac{(\ln t)^2}{2} - \frac{1}{2}(\ln 3)^2\right) = \infty.
\end{aligned}$$

can do by substitution also. let $u = \ln x$

By the Integral Test, the series $\displaystyle\sum_{n=3}^{\infty} \frac{\ln n}{n}$ is divergent. Hence, the series $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n}$ is divergent.

adding first few terms doesn't change converges or diverges.

Example 5.3. Let $p > 0$. Determine whether the following series is convergent or divergent.

$$\sum_{n=3}^{\infty} \frac{1}{n(\ln n)^p}.$$

<u>*Solution.*</u> *Give it a try!* Hint: Substitute $u = \ln u$. Consider different values of p: $p = 1$, $0 < p < 1$ and $p > 1$.

$f(x) = \frac{1}{x(\ln x)^p}$ is positive, cont, decreasing.

$f'(x) = x^{-p}(-p)(\ln x)^{-p-1} + (\ln x)^{-p}(-x^{-2}) < 0$ on $[3, \infty)$

In view of Integral Test:

$$\int_3^\infty \frac{1}{x(\ln x)^p}\,dx \quad \text{let } u = \ln x,\ du = \tfrac{1}{x}\,dx$$

$$= \int_{\ln 3}^{\infty} \frac{1}{u^p}\,du$$

$$= \lim_{t\to\infty} \int_{\ln 3}^{t} u^{-p}\,du \quad \text{— } (*)$$

$p = 1$

$(*) = \lim_{t\to\infty} [\ln|u|]_{\ln 3}^{t}$

$= +\infty$

$p \neq 1$

$(*) = \lim_{t\to\infty} \left[\frac{u^{-p+1}}{-p+1}\right]_{\ln 3}^{t}$

$= \begin{cases} \infty & -p+1 > 0 \\ \text{converges} & -p+1 < 0 \end{cases}$

In summary: $\sum \frac{1}{n(\ln n)^p}$ converges iff $-p+1 < 0 \Rightarrow p > 1$.

5.2 The Comparison Tests

Reference: Stewart's Calculus 8E, Section 11.4

Goal: If we have a series whose terms are "smaller" than those of a known convergent series, then our series is also convergent. If we have a series whose terms are "bigger" than those of a known divergent series, then our series too is divergent.

Direct Comparison Test

Theorem 3. (The Comparison Test)

Suppose that $\sum a_n$ and $\sum b_n$ are series with **positive** terms.

(i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum a_n$ is convergent.

(ii) If $\sum b_n$ diverges to ∞ and $a_n \geq b_n$ for all n, then $\sum a_n$ diverges to ∞.

If bigger series converges, smaller series converges

if smaller series diverges to too, bigger series diverges to too

Proof. Let

$$s_k = \sum_{n=1}^{k} a_n, \quad t_k = \sum_{n=1}^{k} b_n$$

note: $\{s_k\}$, $\{t_k\}$ are increasing sequence because the terms in the series are always positive

denote the k-th partial sums of the series $\sum a_n$ and $\sum b_n$ respectively.

(i) Suppose $\sum b_n$ is convergent, and $a_n \leq b_n$ for all n.

The sequence $\{s_k\}$ and $\{t_k\}$ are increasing, since each term a_n and b_n are positive. Since $a_n \leq b_n$ for all n, we have

$$s_k = \sum_{i=1}^{k} a_n \leq \sum_{n=1}^{k} b_n = t_k.$$

But $\sum b_n$ is convergent, i.e. $\lim_{k\to\infty} t_k = t$ for some $t \in \mathbb{R}$. This implies that $t_k \leq t$ for all k (since $\{t_k\}$ is increasing). So

$$s_k \leq t_k \leq t.$$

We have shown that $\{s_k\}$ is increasing and bounded. By the Monotonic Sequence Theorem, $\lim_{k\to\infty} s_k$ exists, i.e.

$$\sum_{n=1}^{\infty} a_n = \lim_{k\to\infty} s_k = s,$$

for some real number s. That is, $\sum a_n$ is convergent.

(ii) Suppose $\sum b_n$ diverges to ∞, and $a_n \geq b_n$ for all n.
Since $a_n \geq b_n$ for all n, we have $\sum_{n=1}^{k} a_n \geq \sum_{n=1}^{k} b_n$

$$s_k \geq t_k.$$

Since $\lim_{k\to\infty} t_k = \infty$, we deduce that $\lim_{k\to\infty} s_k \geq \lim_{k\to\infty} t_k$

$$\lim_{k\to\infty} s_k = \infty,$$

that is the series $\sum a_n$ also diverges to ∞.

□

Example 5.4. Determine whether the series

$$\sum_{n=1}^{\infty} \frac{100}{2n^2+5n+4} \quad \text{is convergent.}$$

Solution. Set $a_n = \dfrac{100}{2n^2+5n+4}$. Note that

$$a_n = \frac{100}{2n^2+5n+4} < \frac{100}{2n^2} = \frac{50}{n^2}.$$

Set $b_n = \dfrac{50}{n^2}$. We know that

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{50}{n^2} = 50 \sum_{n=1}^{\infty} \frac{1}{n^2}$$

is convergent since $\sum \dfrac{1}{n^2}$ is a p-series with $p > 1$.

Therefore, by the Comparison Test, the series $\sum_{n=1}^{\infty} a_n$ is convergent. □

$\sum \frac{1}{n^p}$ converges iff $p>1$

Bonus:

Determine convergence/divergence of

$\sum_{n=1}^{\infty} \frac{2^n+3^n}{4^n+5^n}$ $\Rightarrow$ can infer it will converge

let $a_n = \frac{2^n+3^n}{4^n+5^n} \le \frac{2^n+3^n}{5^n} \le \frac{3^n+3^n}{5^n} = \frac{2\cdot 3^n}{5^n}$

$\sum b_n = \sum_{n=1}^{\infty} 2\cdot\left(\frac{3}{5}\right)^n$ converges ($\because$ geometric series where $r=\frac{3}{5}$),

by comparison test, $\sum a_n$ converges.

Example 5.5. Test the series $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n}$ for convergence.

Solution. We have tested this series previously using the Integral Test (see Example 5.2). Now, let's use the Comparison Test.

Note that
$$\frac{\ln n}{n} > \frac{1}{n} \quad \text{for all} \quad n \ge 3.$$

We know that $\displaystyle\sum_{n=3}^{\infty} \frac{1}{n}$ is divergent since this is a p-series with $p = 1$ (this is the harmonic series).

So, by the Comparison Test, the series $\displaystyle\sum_{n=3}^{\infty} \frac{\ln n}{n}$ is divergent. Therefore, the series $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n}$ is divergent. $\square$

Sometimes, the following limit version of Comparison Test is easier to apply.

Theorem 4. (The Limit Comparison Test)

Suppose that $\sum a_n$ and $\sum b_n$ are series with **positive** terms.

(i) If $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$ then either both series converge or both series diverge to ∞.

(ii) If $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

(iii) If $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Proof. We first prove part (i).

Let m and M be positive numbers such that $m < c < M$. Since $\dfrac{a_n}{b_n}$ converges to c, for large enough n, we would have
$$m < \frac{a_n}{b_n} < M, \quad \text{for all } n > N.$$

Thus,

$$mb_n < a_n < Mb_n, \quad \text{for all } n > N.$$

We now apply the Comparison Test:

- If $\sum b_n$ converges, then so does $\sum Mb_n$. By part (i) of the Comparison Test, $\sum a_n$ is convergent since $a_n < Mb_n$.
- If $\sum a_n$ converges, then by part (i) of the Comparison Test, $\sum mb_n$ converges since $mb_n < a_n$. This implies that $\sum b_n$ also converges.

Thus, we have proved that $\sum a_n$ converges if and only if $\sum b_n$ converges.

- If $\sum b_n$ diverges to ∞, then so does $\sum mb_n$. By part (ii) of the Comparison Test, $\sum a_n$ diverges to ∞ since $a_n > mb_n$.
- If $\sum a_n$ diverges to ∞, then by part (ii) of the Comparison Test, $\sum Mb_n$ diverges to ∞ since $Mb_n > a_n$. This implies that $\sum b_n$ also diverges to ∞.

Thus, we have proved that $\sum a_n$ diverges to ∞ if and only if $\sum b_n$ diverges to ∞.

The proof of part (ii) is similar. Let M be a positive number. Since $\lim_{n\to\infty} \dfrac{a_n}{b_n} = 0$, there exists N such that

$$0 < \frac{a_n}{b_n} < M, \quad \text{for all } \ n > N.$$

Thus.

$$0 < a_n < Mb_n, \quad \text{for all } \ n > N.$$

Since $\sum b_n$ converges, then so does $\sum Mb_n$. So by part (i) of the Comparison Test, $\sum a_n$ is convergent.

The proof of part (iii) is left as an exercise in the tutorial.

□

When applying the Limit Comparison Test,

- the series $\sum a_n$ is the series whose convergence we want to determine;
- the series $\sum b_n$ is our 'test series' whose convergence is already known to us.

Example 5.6. Test the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ for convergence.

<u>*Solution.*</u> We use the Limit Comparison Test with

$$a_n = \frac{1}{2^n - 1}, \quad b_n = \frac{1}{2^n}.$$

We have

$$\begin{aligned} \lim_{n\to\infty} \frac{a_n}{b_n} &= \lim_{n\to\infty} \frac{1/(2^n - 1)}{1/2^n} \\ &= \lim_{n\to\infty} \frac{2^n}{2^n - 1} \\ &= \lim_{n\to\infty} \frac{1}{1 - 1/2^n} = 1 > 0. \end{aligned}$$

Since $\sum 1/2^n$ is convergent (it is a convergent geometric series), the series $\sum 1/(2^n - 1)$ is also convergent by the Limit Comparison Test. □

Example 5.7. Determine whether the series

$$\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{5 + n^5}}$$

converges or diverges.

<u>*Solution.*</u> The dominant part of the numerator is $2n^2$, and the dominant part of the denominator is $\sqrt{n^5} = n^{5/2}$.

This suggests taking

$$a_n = \frac{2n^2 + 3n}{\sqrt{5 + n^5}}, \quad b_n = \frac{2n^2}{n^{5/2}} = \frac{2}{n^{1/2}}.$$

In view of Limit Comparison Test, compute $\lim_{n\to\infty} \frac{a_n}{b_n}$

$$
\begin{aligned}
\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{2n^2+3n}{\sqrt{5+n^5}}\cdot\frac{n^{1/2}}{2} \\
&= \lim_{n\to\infty}\frac{2n^{5/2}+3n^{3/2}}{2\sqrt{5+n^5}} \quad \div \frac{n^{5/2}}{n^{5/2}} \\
&= \lim_{n\to\infty}\frac{2+\frac{3}{n}}{2\sqrt{\frac{5}{n^5}+1}} \\
&= \frac{2+0}{2\sqrt{0+1}} = 1.
\end{aligned}
$$

Notice that $\sum b_n$ is divergent since it is the p-series with $p = \frac{1}{2} < 1$. By the Limit Comparison Test, the series $\sum a_n$ is divergent. $\square$

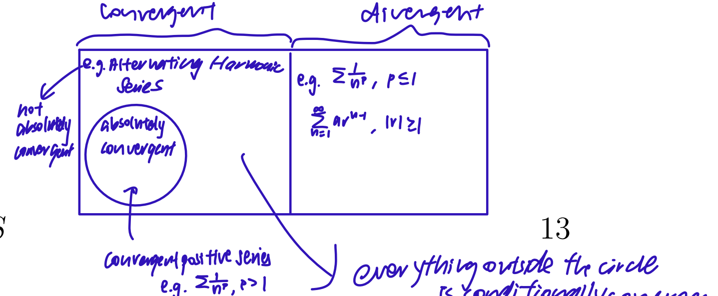
Alternatively,

$$a_n = \frac{2n^2+3n}{\sqrt{5+n^5}} \geq \frac{2n^2}{\sqrt{5+n^5}} \geq \frac{2n^2}{\sqrt{5n^5+n^5}} = \frac{2n^2}{\sqrt{6}\,n^{5/2}} = \frac{2}{\sqrt{6}\sqrt{n}} = b_n$$

Since $\sum b_n = \sum \frac{2}{\sqrt{6}\sqrt{n}} = \frac{2}{\sqrt{6}} \sum \frac{1}{n^{1/2}}$ p-series

diverges (p-series, $p = \frac{1}{2} \leq 1$),

by comparison test, $\sum a_n$ also diverges (to $+\infty$)



5.3 Absolute and Conditional Convergence

Reference: Stewart's Calculus 8E, Section 11.5, 11.6

> **Goal:** Understand the difference between absolute convergence and conditional convergence. Apply the Alternating Series Test to determine the convergence of an alternating series.

for series with positive terms, there's no difference between the two

In the previous sections, we studied the convergence of series with positive terms, but we still lack the tools to analyze series with both positive and negative terms such as

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$$

Such a series is called an **alternating series** which has the following form:

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n \quad = a_1 - a_2 + a_3 - a_4 + \ldots$$

where $\{a_n\}$ is a sequence with positive terms. $\sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + \ldots$

A key concept in understanding these type of series is the concept of absolute convergence.

Definition 1. (Absolute Convergence)

A series $\sum a_n$ is called **absolutely convergent** if the series $\sum |a_n|$ converges.

For positive series ie. $\sum a_n$, $a_n > 0$, $\sum |a_n| = \sum a_n \Rightarrow$ absolute convergence = convergence

Example 5.8. The series

$$\sum a_n = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$$

is absolutely convergent because the corresponding series with absolute values is a convergent p-series:

$$\sum |a_n| \quad \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$$

$$= \sum \frac{1}{n^2} \quad (p = 2 > 1)$$

$\therefore$ converges $\Rightarrow$ p-series

Eg.
$\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots$
Alternating Harmonic Series.
$\sum |a_n| = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots$ divergent
So the Alternating Harmonic Series is NOT absolutely convergent.

The following result states that if the series of absolute values converges then the original series converges.

Theorem 5. (Absolute Convergence implies Convergence)

If $\sum a_n$ is absolutely convergent, then $\sum a_n$ converges. → converse is not true

Proof. We have

$$0 \le \overbrace{a_n + |a_n|}^{\text{non-negative}} \le 2|a_n|.$$

→ apply comparison test

write $a_n = a_n + |a_n| - |a_n|$

Thus $\sum(a_n + |a_n|)$ is a series with positive terms (dropping all 0's).

$a_n + |a_n| = \begin{cases} 0 & \text{if } a_n \le 0 \\ 2|a_n| & \text{if } a_n > 0 \end{cases}$

Suppose $\sum a_n$ is absolutely convergent, i.e. $\sum |a_n|$ converges. Then $\sum 2|a_n|$ is convergent. By the Comparison Test, the series $\sum(a_n + |a_n|)$ is convergent.

Our original series is the difference of two convergent series and hence it converges:

$$\sum a_n = \sum((a_n + |a_n|) - |a_n|) = \underbrace{\sum(a_n + |a_n|)}_{\text{convergent}} - \underbrace{\sum |a_n|}_{\text{convergent}}.$$

□

Bonus:

$\sum_{n=1}^{\infty} n\sin\left(\frac{1}{n^2}\right)$

let $a_n = n\sin\left(\frac{1}{n^2}\right)$

choose $b_n = \frac{1}{n}$

$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{n\sin\left(\frac{1}{n^2}\right)}{\frac{1}{n}}$

$= \lim_{n\to\infty} \frac{\sin\left(\frac{1}{n^2}\right)}{\frac{1}{n^2}}$

$= \lim_{x\to\infty} \frac{\sin(x)}{x}$ L'Hopital Rule

$= \lim_{x\to\infty} \frac{\cos x}{1}$

$= 1 > 0$

Since $\sum b_n = \sum \frac{1}{n}$ diverges by Limit Comparison Test, $\sum_{n=1}^{\infty} n\sin\left(\frac{1}{n^2}\right)$ diverges

$\boxed{\textbf{Example 5.9.}}$ Determine whether the series

$$\sum_{n=1}^{\infty} \frac{\cos n}{n^2} = \frac{\cos 1}{1^2} + \frac{\cos 2}{2^2} + \frac{\cos 3}{3^2} + \cdots$$

is convergent or divergent.

Solution. The series has both positive and negative terms, but it is not alternating: The first term is positive, the next three are negative etc (in fact, the signs change irregularly).

The series of absolute values is

cos n can be positive or negative

$$\sum_{n=1}^{\infty} \left|\frac{\cos n}{n^2}\right| = \sum_{n=1}^{\infty} \frac{|\cos n|}{n^2}.$$

Since $|\cos n| \leq 1$ for all n, we have

$$\frac{|\cos n|}{n^2} \leq \frac{1}{n^2}.$$

We know that $\sum \frac{1}{n^2}$ is convergent (p-series with $p = 2$). Therefore, by the Comparison Test, the series $\sum \frac{|\cos n|}{n^2}$ is convergent.

Hence, the series $\sum \frac{\cos n}{n^2}$ is absolutely convergent. By Theorem 5, the series $\sum \frac{\cos n}{n^2}$ is convergent. $\square$

Example 5.10. Does the series $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$ converge absolutely?

<u>*Solution.*</u> We apply the Integral Test to the series of absolute values: $\displaystyle\sum_{n=2}^{\infty} \left|\frac{(-1)^n}{n \ln n}\right| = \sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

Let $f(x) = \frac{1}{x \ln x}$ positive, continuous, and decreasing on $[2, \infty)$.

Using the substitution $u = \ln x$, $du = x^{-1}\, dx$, we have

In view of Integral Test

$$\begin{aligned} \int_2^{\infty} \frac{1}{x \ln x}\, dx &= \int_{\ln 2}^{\infty} \frac{1}{u}\, du \\ &= \lim_{t \to \infty} \int_{\ln 2}^{t} \frac{1}{u}\, du \\ &= \lim_{t \to \infty} (\ln t - \ln(\ln 2)) \\ &= \infty. \end{aligned}$$

Therefore, $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ diverges, and so $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$ is not absolutely convergent. $\square$

NOTE: In the preceding example, we showed that the series $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$ does not converge absolutely, but we still do not know whether or not it converges. In general, a series which is not absolutely convergent may or may not be convergent. In the case when it converges (without absolute convergence), we say the the series is conditionally convergent.

Definition 2. (Conditional Convergence)

A series $\sum a_n$ is called **conditionally convergent** if it converges but $\sum |a_n|$ diverges.

not absolutely convergent

If we encounter a series that is not absolutely convergent, how can we determine whether it is conditionally convergent? The Integral Test and Comparison Test cannot be used because they apply only to series with **positive** terms.

Our next test applies to alternating series. The proof is beyond the scope of our study.

→ just use without proof.

Theorem 6. (Alternating Series Test)

Let $\{a_n\}$ be a decreasing positive sequence that converges to 0:

$$a_1 \geq a_2 \geq a_3 \geq a_4 \geq \cdots \geq 0, \quad \text{and} \quad \lim_{n\to\infty} a_n = 0.$$

Then the following alternating series converges:

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n = a_1 - a_2 + a_3 - a_4 + \cdots$$

Note: The alternating series above starts with a positive term and alternates between positive and negative terms. The result also applies if it starts with a negative term, and alternates between positive and negative terms.

Example 5.11. Show that $S = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ is conditionally convergent.

① Check absolute convergence

② Check convergence.

Solution. The terms $a_n = \frac{1}{\sqrt{n}}$ form a decreasing sequence that converges to 0. The Alternating Series Test implies that $S = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ is convergent.

However, S is only conditionally convergent because the series of absolute values $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{\sqrt{n}} \right| = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is a divergent p-series (since $p < 1/2$). □

Example 5.12. Is the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3+1}$ convergent?

Solution. Let $a_n = \frac{n^2}{n^3+1}$. In view of the Alternating Series Test, we want to show that $\{a_n\}_{n=1}^{\infty}$ is decreasing and $a_n \to 0$.

It is not obvious that the sequence $a_n = \frac{n^2}{n^3+1}$ is decreasing. Let's consider the related function $f(x) = \frac{x^2}{x^3+1}$, where $f(n) = a_n$. Its derivative is

$$f'(x) = \frac{2x(x^3+1) - x^2(3x^2)}{(x^3+1)^2} = \frac{x(2-x^3)}{(x^3+1)^2}.$$

Since we are considering only positive x, we see that

$$f'(x) < 0 \iff 2 - x^3 < 0 \iff x > 2^{1/3}.$$

Thus, f is (strictly) decreasing on the interval $(2^{1/3}, \infty)$. This means that $f(n+1) < f(n)$ and therefore $a_{n+1} < a_n$ for all $n \geq 2$. On the other hand, it is clear that

$$a_2 = \frac{2^2}{2^3+1} = \frac{4}{9} < \frac{1}{2} = a_1.$$

So the sequence $\{a_n\}_{n=1}^{\infty}$ is decreasing.

The limit of $\{a_n\}$ is readily verified:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n^2}{n^3+1} = \lim_{n\to\infty} \frac{1/n}{1+1/n^3} = 0.$$

Hence, the given series is convergent by the Alternating Series Test. □

Example 5.13. Is the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{n^3+1}$ conditionally convergent or absolutely convergent?

Solution. *Give it a try!*

For this, Check $\sum_{n=1}^{\infty}\left|(-1)^{n+1}\frac{n^2}{n^3+1}\right|$

$$= \sum_{n=1}^{\infty}\frac{n^2}{n^3+1}$$

$$\frac{n^2}{n^3+1} \geq \frac{n^2}{n^3+n^3} = \frac{n^2}{2n^3} = \frac{1}{2n}$$

Since $\sum\frac{1}{2n}$ diverges (p-series, $p=1$)
by comparison Test, the "bigger" series $\sum\frac{n^2}{n^3+1}$ diverges.

So $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{n^3+1}$ is NOT absolutely convergent

So by the previous question, the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{n^3+1}$ is

conditionally convergent.

Alternatively,

$$\int_2^{\infty}\frac{x^2}{x^3+1}\,dx = \lim_{t\to\infty}\frac{1}{3}\Big[\ln|x^3+1|\Big]_2^t$$

$$= \lim_{t\to\infty}\frac{1}{3}\left(\ln|t^3+1| - \ln|9|\right)$$

$$= +\infty$$

By Integral Test, $\sum_{n=2}^{\infty}\frac{n^2}{n^3+1}$ diverges like a geometric series

5.4 The Ratio and Root Tests

Reference: Stewart's Calculus 8E, Section 11.6

Goal: The convergence tests developed so far cannot be easily applied to series containing factorial terms or n-th powers. For this, we will need the Ratio Test and Root Test, which are also of key importance in the study of power series in the next chapter.

Theorem 7. (Ratio Test)

Let $\{a_n\}$ be a sequence and assume that the following limit exists:

$$\rho = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|.$$

(i) If $\rho < 1$, then $\sum a_n$ converges absolutely.

(ii) If $\rho > 1$ or $\rho = \infty$, then $\sum a_n$ diverges.

like geometric series

(iii) If $\rho = 1$, then Ratio Test is inconclusive (the series may converge or diverge).

Proof.

Case (i) $\rho < 1$.

Choose number r such that $\rho < r < 1$.

Since $\left|\frac{a_{n+1}}{a_n}\right|$ converges to ρ, there exists a number N such that

$$\left|\frac{a_{n+1}}{a_n}\right| < r \quad \text{for all} \quad n \geq N.$$

i.e.

$$|a_{n+1}| < r|a_n| \quad \text{for all} \quad n < N.$$

In particular, we have

$$\begin{aligned}
|a_{N+1}| &< r|a_N| \\
|a_{N+2}| &< r|a_{N+1}| < r^2|a_N| \\
&\vdots \\
|a_{N+k}| &< r^k\underbrace{|a_N|}_{\text{constant}}.
\end{aligned}$$

Therefore,

$$\begin{aligned}\sum_{n=N}^{\infty}|a_n| &= \sum_{k=0}^{\infty}|a_{N+k}| \\ &\leq \sum_{k=0}^{\infty} r^k|a_N| \\ &= |a_N|\sum_{k=0}^{\infty} r^k.\end{aligned}$$

The geometric series on the right-hand side converges because $0 < r < 1$, and so $\sum_{n=N}^{\infty}|a_n|$ converges by the Comparison Test. Thus, the series $\sum a_n$ converges absolutely.

Case (ii) $\rho > 1$ or $\rho = \infty$.

Choose a number r such that $1 < r < \rho$. Arguing as before with inequalities reversed, there is an integer N such that

$$|a_{N+k}| \geq r^k|a_N|$$

for all $k \geq 0$.

Since r^k tends to ∞ (because $r > 1$), we see that the term a_{N+k} do not tend to 0. By the nth Term Test for Divergence, we conclude that the series $\sum a_n$ is divergent.

Case (iii) $\rho = 1$.

Example (5.14) below show that when $\rho = 1$, both convergence and divergence is possible, so the test is inconclusive in this case. □

<u>Summary of Ratio Test:</u>

Whenever ratio test works:

- absolute convergence $\rho < 1$
- divergence $\rho > 1, +\infty$

$\rho = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$

Example 5.14.

Consider the series $\sum_{n=1}^{\infty} n^2$ and $\sum_{n=1}^{\infty} n^{-2}$. Show that for both series, we have $\rho = 1$, and so the Ratio Test is inconclusive.

Solution. Let $a_n = n^2$. Then

$$\begin{aligned}
\rho &= \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| \\
&= \lim_{n\to\infty} \frac{(n+1)^2}{n^2} \\
&= \lim_{n\to\infty} \frac{n^2+2n+1}{n^2} \\
&= \lim_{n\to\infty} \left(1 + \frac{2}{n} + \frac{1}{n^2} \right) \\
&= 1.
\end{aligned}$$

On the other hand, let $b_n = n^{-2}$. Then

$$\begin{aligned}
\rho &= \lim_{n\to\infty} \left| \frac{b_{n+1}}{b_n} \right| \\
&= \lim_{n\to\infty} \frac{n^2}{(n+1)^2} \\
&= \lim_{n\to\infty} \frac{n^2}{n^2+2n+1} \\
&= \lim_{n\to\infty} \frac{1}{1+\frac{2}{n}+\frac{1}{n^2}} \\
&= 1.
\end{aligned}$$

Thus, $\rho = 1$ in both cases. However, $\sum_{n=1}^{\infty} n^2$ diverges (since the n-th term does not vanish), whereas $\sum_{n=1}^{\infty} n^{-2}$ converges (since it is a p-series with $p = 2$).

This shows that both convergence and divergence are possible when $\rho = 1$. □

Example 5.15. Using the Ratio Test, show that the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n!}$ converges.

Solution. We compute the limit ρ. Let $a_n = \dfrac{1}{n!}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{1}{(n+1)!} \cdot \frac{n!}{1} = \frac{1}{n+1}.$$

$$\rho = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{1}{n+1} = 0.$$

Since $\rho < 1$, the series converges by the Ratio Test. □

Example 5.16. Apply the Ratio Test to determine if $\displaystyle\sum_{n=1}^{\infty}\frac{n^2}{2^n}$ converges.

Solution. Let $a_n = \dfrac{n^2}{2^n}$. We have

$$\begin{aligned}
\frac{a_{n+1}}{a_n} &= \frac{(n+1)^2}{2^{n+1}}\cdot\frac{2^n}{n^2} \\
&= \frac{1}{2}\left(\frac{n^2+2n+1}{n^2}\right) \\
&= \frac{1}{2}\left(1+\frac{2}{n}+\frac{1}{n^2}\right)
\end{aligned}$$

$$\rho = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{1}{2}\left(1+\frac{2}{n}+\frac{1}{n^2}\right) = \frac{1}{2}.$$

Since $\rho < 1$, the series converges by the Ratio Test. □

Example 5.17. Determine whether $\displaystyle\sum_{n=0}^{\infty}(-1)^n\frac{n!}{100^n}$ converges.

Solution. Let $a_n = (-1)^n\dfrac{n!}{100^n}$. Then

removes alternating sign

disappears cus of absolute function.

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)!}{100^{n+1}}\cdot\frac{100^n}{n!} = \frac{n+1}{100}.$$

The ratio diverges to infinity:

$$\rho = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n+1}{100} = \infty.$$

The series diverges by the Ratio Test. □

For some series, it is more convenient to use the following Root Test, based on the limit of the nth roots $\sqrt[n]{a_n}$ rather than the ratios a_{n+1}/a_n. The proof of the Root Test, like that of the Ratio Test, is based on a comparison with geometric series (see exercise in the tutorial).

proof in tutorial

Theorem 8. (Root Test)

Let $\{a_n\}$ be a sequence and assume that the following limit exists:

$$L = \lim_{n\to\infty} \sqrt[n]{|a_n|}.$$

we root test when $a_n = (\ldots n)^n$

(i) If $L < 1$, then $\sum a_n$ converges absolutely.

(ii) If $L > 1$ or $L = \infty$, then $\sum a_n$ diverges.

(iii) If $L = 1$, the Root Test is inconclusive. The series may converge or diverge.

Example 5.18. Determine whether $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n}{2n+3}\right)^n$ converges.

Solution. Let $a_n = \left(\dfrac{n}{2n+3}\right)^n$. Then

$$L = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{n}{2n+3} = \frac{1}{2}.$$

Since $L < 1$, the series converges by the Root Test. $\square$

Example 5.19.

(a) Prove that $\lim_{n\to\infty} \frac{\ln n}{n} = 0$.

(b) Prove that $\lim_{n\to\infty} n^{1/n} = 1$.

(c) Determine the convergence/divergence of the series $\sum_{n=1}^{\infty} a_n$, where

$$a_n = \begin{cases} \dfrac{n}{2^n} & \text{if } n \text{ is odd} \\ \dfrac{1}{2^n} & \text{if } n \text{ is even} \end{cases}$$

Solution. *Give it a try!*

a) $\lim_{n\to\infty} \frac{\ln n}{n} = \lim_{n\to\infty} \frac{\frac{1}{n}}{1} = 0$

b) $\lim_{n\to\infty} n^{\frac{1}{n}} = \lim_{n\to\infty} e^{\ln n^{\frac{1}{n}}} = \lim_{n\to\infty} e^{\frac{1}{n}\ln n} = e^{\lim_{n\to\infty} \frac{1}{n}\ln n} = e^0 = 1$

c) $\sum_{n=1}^{\infty} a_n = \frac{1}{2} + \frac{1}{2^2} + \frac{3}{2^3} + \frac{1}{2^4} + \frac{5}{2^5} + \frac{1}{2^6} + \ldots$

In view of Root Test:

$L = \lim_{n\to\infty} |a_n|^{\frac{1}{n}}$

$\frac{1}{2^n} \le a_n \le \frac{n}{2^n}$

can drop absolute signs since all the terms are positive

$\frac{1}{2} = \left(\frac{1}{2^n}\right)^{\frac{1}{n}} \le |a_n|^{\frac{1}{n}} \le \left(\frac{n}{2^n}\right)^{\frac{1}{n}} = \frac{n^{\frac{1}{n}}}{2}$

Since $\lim_{n\to\infty} \frac{n^{\frac{1}{n}}}{2} = \frac{1}{2}$ (by part (b)),

By squeeze, $L = \lim_{n\to\infty} |a_n|^{\frac{1}{n}} = \frac{1}{2} < 1$

By Root Test, $\sum a_n$ converges.